

Alok Kumar Mishra

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EDUCATION

- Indian Institute of Information Technology, Jabalpur** Madhya Pradesh, India
Bachelor of Technology - Electronics & Communication Engineering Aug 2024 – Present

EXPERIENCE

- DigiLocker (MeitY), Government of India** Remote
Security Contributor (AppSec) Feb 2025 – Apr 2025
 - Application Security & Android RE:** Responsibly disclosed **10+ P1/P2 vulnerabilities** in DigiLocker's Android app (**500M+ users**) via dynamic/static analysis using **Frida, JEB, Radare2**.
 - Remediation & Recognition:** Worked with DigiLocker engineers on fixes through detailed reports; received **official Govt. of India acknowledgment** for responsible disclosure.
- Rizin Project (Open-Source RE Framework)** Remote
Open Source Contributor – RE Tooling Nov 2025 – Mar 2026
 - Analysis Layer Improvements:** Implemented **ahie** commands for enum immediate rendering in disassembly with full hint storage, serialization, and project persistence.
 - Tooling & Utilities:** Added **rz-ar** utility for robust archive extraction with Windows path handling; enhanced **rz-find** with multi-path support.
 - Core Stability & Hardening:** Fixed endless loop in shell history, added ELF versym bounds checking, and unified address description API across core modules.
 - Ecosystem Modernization:** Led Cap'n Proto migration in rz-silhouette (client + server) for better performance and memory safety; redesigned rizin-notebook with SQLite architecture and safe rendering.

PROJECTS

- RzWeb – Rizin on Web (WebAssembly Reverse Engineering Platform)** GitHub
WebAssembly (WASM), WASI, TypeScript, Python, Shell, Emscripten
 - Built **RzWeb**, a complete browser-native reverse engineering platform by compiling Rizin to WebAssembly (WASM), enabling secure static analysis directly in the browser.
 - Engineered sandbox-safe WASI bindings and core RE workflows (disassembly, interactive console, graph navigation) with responsive TypeScript UI, optimized for instant loading on cached binaries.
 - Shipped production-grade deployment achieving **500+ stars**, **50+ forks**, and **100K+ visits** in the first week, reducing setup friction from hours to under 2 minutes.
- Arthyx – Quant Finance Document Intelligence** GitHub
Next.js, TypeScript, Neo4j, Vector Search
 - Engineered a **RAG-based quant system** for deep analysis of **500+ page financial PDFs** (Basel III, NPA, CRAR).
 - Modeled financial entities and risk propagation using **Neo4j knowledge graphs**, reducing manual review by **40%**.
- HexaCiphers – Social Media Intelligence & OSINT** GitHub
React, Python, NLP
 - Built a real-time multilingual OSINT platform for Twitter/X campaign and narrative analysis.
 - Implemented bot detection and sentiment pipelines achieving **80% accuracy** for cyber threat intelligence use cases.

TECHNICAL SKILLS

- Languages:** Python, C/C++, Java, JavaScript, TypeScript, Bash, SQL, Smali, HTML/CSS
- Cybersecurity / RE:** VAPT, Bug Bounty, AppSec, Android RE, Frida, Ghidra, IDA Pro, JEB, Radare2, Burp, Metasploit
- Web / Backend:** React, Next.js, Node.js, FastAPI, Firebase, Tailwind, REST APIs
- AI / Finance:** RAG Systems, NLP, Vector Search, Neo4j, Gemini API, Basel III, NPA/CRAR
- Tools:** Git, Docker, Linux, Android Studio, Postman, Vercel

ACHIEVEMENTS

- Acknowledged **Security Contributor** by DigiLocker (GOI) for 10+ critical vulnerabilities (500M+ users).
- 1st Place** – CanYouHackIt 2025 (Institute's Flagship Student-Led Hackathon, 200+ students), Team Lead.
- Top 20 / 700+ teams Nationally** – HackTheFuture 2025, IIT Gandhinagar.
- JEE Main Top 1.2%**; CBSE Merit (**100/100 CS**); **Global Rank 385** in CodeChef Starters 176C.

VOLUNTEER EXPERIENCE

- Jagrati – Non-Profit Education Initiative** Jabalpur, India
Mathematics Tutor (Volunteer) Aug 2024 – Present
 - STEM Education Outreach:** Provided free mathematics tutoring to underprivileged students preparing for Navodaya entrance exams.
- TechFest “AbhiKalpan” – IIITDM Jabalpur** Jabalpur, India
Student Volunteer Apr 2025
 - Event Operations:** Assisted in planning and on-site coordination of institute's annual technical festival.